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TEWL Measurement Device Technical Memorandum 2026

Background

Transdermal drug delivery (TDD) is the process of applying a patch, cream, or gel to the skin to deliver medication to patients at a specific dosage. TDD is desirable because it is minimally invasive compared to oral and parenteral routes and avoids first pass loss in the liver allowing for dosing to be more accurate. On a microscopic level, transdermal drug delivery requires the drug to permeate through the stratum corneum (SC)–the outermost layer of the epidermis that functions as the skin's barrier against toxins and pathogens–in order to enter the bloodstream. This barrier's structure consists of tightly-packed corneocytes that result in low permeability and high structural resistance. TDD is achieved when the drugs pass through the stratum corneum transcellularly To develop more effective methods for transdermal drug delivery, a deeper understanding of the mechanical properties of the SC is required.

To characterize the mechanical properties of the SC barrier function, the team is performing various tests on the skin under different applied strains and comparing fractional viscoelastic parameters and TEWL measurements to note any significant change. It is important to perform *in vivo* skin-stretching tests because the structure and geometry of the SC can change drastically when the skin is being stretched, which may alter its permeability. During the testing period, the team will be recording transepidermal water loss (TEWL) measurements at various strains. TEWL is an indicator of how intact the barrier function of the skin is, as it measures the flux of water vapor escaping from the SC. Thus, it can provide insight into the skin's ability to retain moisture, uphold barrier function, and the rate of drug transport through the SC. TEWL-measuring devices can measure the skin barrier function by assessing time-dependent water evaporation. The team will use these devices to test the hypothesis that stretching the skin will increase TEWL values and thus increase permeability.

TEWL-measuring devices that are currently available on the market include the Vapometer, Tewameter, and the AquaFlux. The Vapometer, developed by Delfin technologies, is a handheld TEWL measuring device that utilizes the closed-chamber principle, eliminating environmental disturbances such as ambient air flow. The Vapometer has been used in numerous publications on skin barrier functionality. The Tewameter is another hand held TEWL measuring device that uses the open-chamber method to record TEWL continuously without influencing the skin's microenvironment. The Tewameter consists of 30 pairs of relative humidity and temperature sensors, allowing for an increased amount of data collected. Another open chamber TEWL measuring device is the AquaFlux, which has also been used in several published studies on the skins barrier and has a recorded study illustrating higher performance in the categories of sensitivity and repeatability. The costs of these TEWL measuring devices are notably high, ranging from $8000 to $12000. Thus, there have been numerous studies on research labs developing their own variation of a TEWL measurement device. In the article WASP: Wearable Analytical Skin Probe for Dynamic Monitoring of Transepidermal Water Loss article, a research team at the University of Michigan developed their own TEWL measuring device for under $500. This custom-built device allowed them to accomplish goals specific to their project, such as being able to take long-term TEWL measurements.

Initial Status

On June 1st, 2026, the team determined that the lab's current TEWL measuring devices, the VapoMeter and Aquaflux, were both unfit for the lab's needs as they often gave incorrect measurements with large margins of error. For example, the following TEWL values were taken in one continuous run: 17.7, 15, 15.3, 15.9, 16.8, 19.5, 19.2, 15, 14.5, 13.8. With the cost of purchasing a new TEWL device estimated to be around upwards of $10,000, the team decided to develop a custom-built TEWL measurement device. Referencing the device in the WASP: Wearable Analytical Skin Probe for Dynamic Monitoring of Transepidermal Water Loss article as the backbone of the design, the team began to develop the device.

Work Done

Circuit Design

The TEWL-measuring device is composed of the following key components: a type-K thermocouple, relative humidity sensor, air pump, microcontroller, 3.7V LiPo battery, and an outer casing to keep all components in a known area. Figure 1 is a schematic of the device's circuitry.

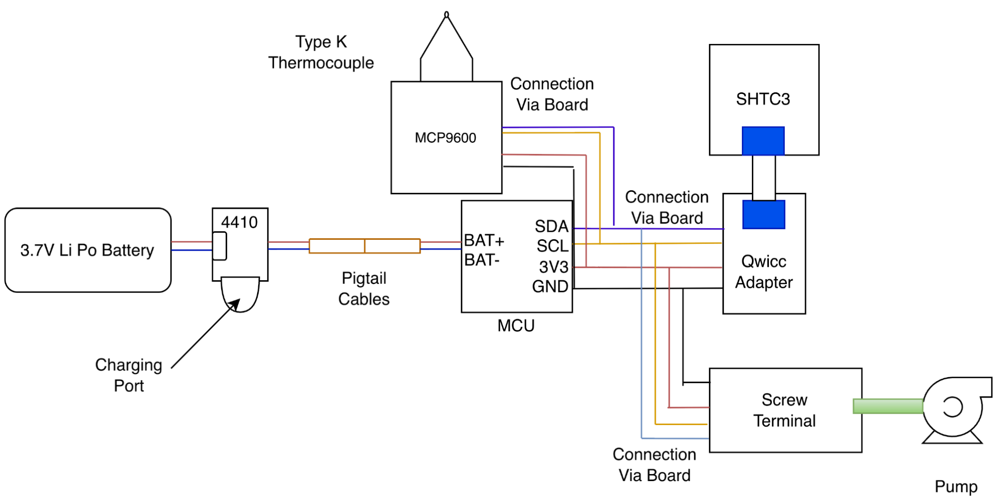


Figure 1: Electrical Schematic of TEWL measuring device

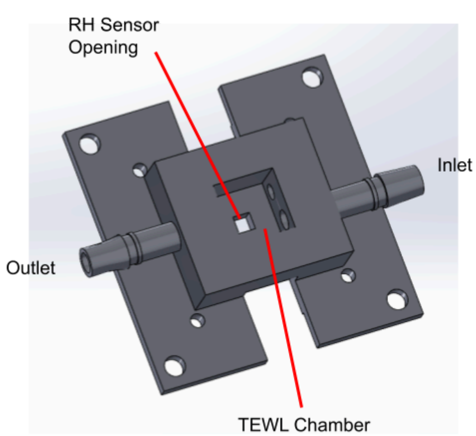
The 3.7 lithium battery is first connected to a cable assembly in accordance with power and ground. The cable can then connect to the 4410 board on the JST port. The female pigtail plug will also be connected to the 4410 board on the 5V and GND pins accordingly. The male pigtail plug is attached to the end of the female pigtail plug and the BAT+ and BAT - on the microcontroller. Note that all components (starting from the microcontroller) will be placed on a protoboard. For easy removal and modifications, components will be attached to the protoboard using male and female pins. The 3V3, GND, SLC, SDA pin from the microcontroller will connect to the thermocouple board. A type-K thermocouple will be screwed within the screw terminals located on the thermocouple board. The microcontroller will connect its 3V3, GND, SLC, SDA pins to the screw terminal adapter. The pump will then be connected to the screw terminal adapter. The pump out is connected to a moisture filter to ensure the majority of the water from the air is filtered prior to entering the chamber. This filter is composed of cotton ball and 4A pellets and resin connectors. The microcontroller GND, 3V3, SLC, SDA, pins will be connected to the Qwicc adapter board. Using the Qwicc cable, the adapter board will connect to the relative humidity (RH) sensor.

The micropump does not produce a strong enough air current to properly dry out the TEWL chamber when attached to the skin. The team tried alternative approaches to flush out the chamber using a ball pump and a hair dryer. The ball pump was attached to the valve through the filter and then hand-pumped, but also did not produce a sufficiently strong air current to flush out the chamber. The hair dryer was able to bring the RH down on the “Warm” blow-dry setting, but also affected the temperature of the chamber, which may bring up additional complications. The “Cool” setting does not seem to affect the temperature as much, but does not dry out the chamber as well as the “Warm” setting.

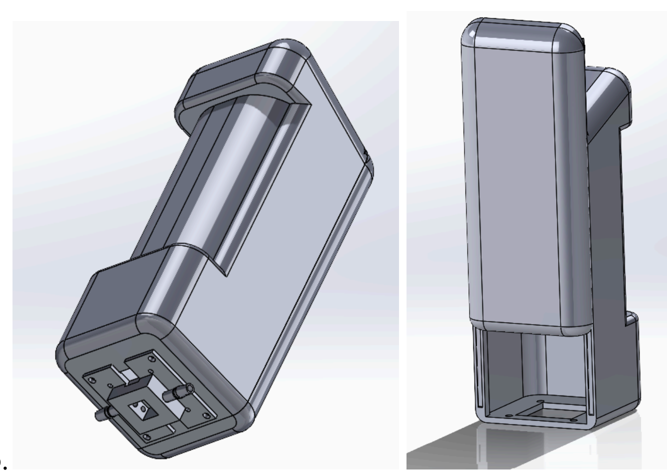
The circuit design in Figure 1 is what the lab found to be the most optimal design. Alternatives to this design are as follows: Removing the pigtail cables and the 4410 boards entirely and directly soldering the battery onto the microcontroller. To charge the battery, the operator can simply plug in a power source into the USB port and monitor the charging LED light on top of the board. The team decided to incorporate the pigtail cables to minimize the width of the design and maximize user friendliness as it allows the team to easily disconnect the power.

Mechanical Design

The sensor measurements are recorded within a resin-printed TEWL chamber, shown in Figure 2a. This chamber has a square opening with dimensions of 8x8mm and is 6mm in depth. The RH sensor’s board is placed flush on the top of the chamber, and the sensor itself can access the chamber through the small square opening on the ceiling. The thermocouple is placed into the chamber through a small hole on one of the sides of the chamber, and is covered with parafilm from the outside to minimize air leakage. There are two one-way valves connected to the chamber. One of the valves is connected to the tube that is connected to the outlet side of the pump. This valve flushes out humidity in the chamber with external dry air and also ensures that no air from the chamber exits the chamber back through the tube. The second valve acts as an outlet for air to leave the chamber during the drying/flushing process. The final component is the filter, which is meant to filter out moisture in the air flowing into the chamber. It is composed of an aluminum rod, 5Å molecular sieves, and glass wool, and then connected to the chamber’s tubing.



a.



b.

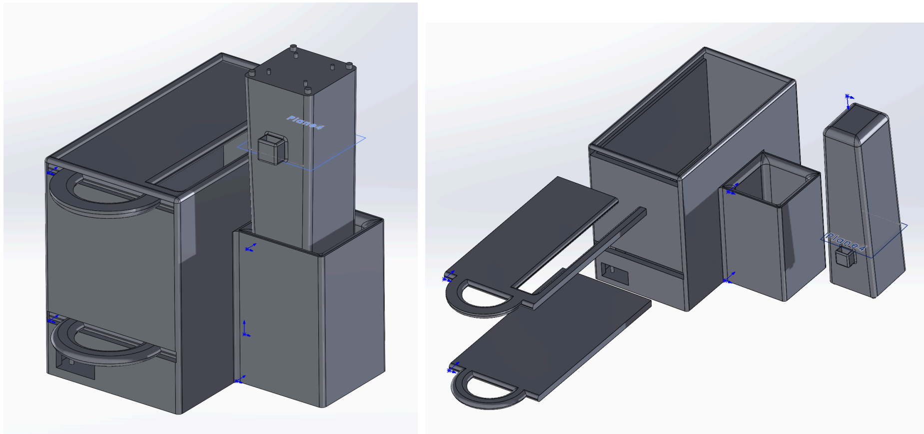
c. 

Figure 2.

a. CAD Model of the resin-printed TEWL chamber.[1]

b. CAD Models of the sliding case design for the TEWL device.

c. CAD Models of the landline-telephone-inspired design for the device.

There are two possible designs for the casing of the device, both of which are to be 3D-printed with PLA. The first design is shown in Figure 2b., which houses all of the components and electronics of the device in one singular body. One of the sides can be removed with a sliding mechanism in order to access the device's components. This design would compile all components into a device with a singular body, but may be bulky and harder to hold for the user.

The second possible design is shown in Figure 2c. It consists of two main compartments: The main body, which will store the majority of the boards and circuitry of the device, and a handheld probe that has the RH sensor and TEWL chamber attached to the bottom. This design would allow the battery to be continuously charged, and more space within the main body if there need to be any additions to the circuitry.

The base manages the TEWL circuitry through a two layer system. The bottom layer is a storage space for the battery and 4410 board. The bottom portion of the main box has two designated holes, one for the output of the pigtail cable and one for easy charging. To ensure that charging the battery using the 4410 board is successful, we stabilized the board with additional supports to keep the charging output pointing towards the hole.

To divide the top and bottom layer, a 3D printed slider was used (Figure 2c). The top layer acts as the storage for all parts attached to the protoboard and the pump. Walls and stabilizers were added to ensure that the protoboard and pump are appropriately stabilized. The cables and tubes that are connected to the chamber and RH sensor will emerge from the slit in the top layer slider.

The probe is a tapered rectangular prism that fits the RH sensor and TEWL chamber on one end. There are three compartments that will be used to separate the cables. One compartment is designated for the qwicc cable and thermocouple, one is for the inlet pump, and one supports the one-way valve.

I2C Programming

[1] All CAD models can be found in the team's Google Drive under ATT > 2026 > Summer 2026 ? Solidworks Modeling > TEWL Measuring Device.

The system uses I2C and digital communication from the Seeeduino XIAO nRF52840 board to perform all of its functions. The <Wire.h> library is used for writing I2C code. More information about I2C code can be found here or in Arduino forums. Both the RH sensor and the thermocouple have additional libraries that make their I2C code easier to write. These are the <Adafruit_SHTC3.h> library and the <Adafruit_MCP9600.h> library (which also includes the MCP9601 library), respectively.

The code uses digital communication to the pump screw terminal adapter, where the I/O X pin is connected to the digital pin 3 and set to one or zero (on at full power or off). Both I2C and UART communication should be possible with the pump. Specifically, there is a library for UART communication to the Xavitech pumps. However, the code from the library doesn't work as it was unable to stop or start the pump.

In addition to the I2C communication, the team has started to setup bluetooth communication so that a computer doesn't have to be physically connected to the TEWL device whenever it is used. So far, the bluetooth communication from the Seeeduino XIAO nRF52840 board only connects to a phone via the LightBlue app under the device connection name "Arduino". The current version of the code sends RH data from the XIAO nRF52840 to the phone in hex values, on request from the phone. Ideally, this code would be able to send data in a constant stream to a computer for easy reading. See Appendix A for instructions on how to use LightBlue to connect to the Arduino.

Table 1 summarizes the current code files for the TEWL reading device below, which are all found in the ATT github repository in the "Bananmeter_Code" folder. As shown in Table 1, code for different sensors has so far been separated for ease of debugging, but should eventually be combined into one file (or header and cpp files) for the functional device.

Name of code file	Name of peripheral device it interacts with	Intended purpose	Working?
bluetooth_test	n/a	Example from https://wiki.seeedstudio.com/XIAO-BLE-Sense-Bluetooth-Usage/ that turns an LED on the seeeduino on/off	Yes
mcu_prototype	RH sensor	To use I2C without the adafruit_shtc3 library to connect to the RH sensor and take readings from it.	No. The I2C doesn't read anything from the RH
pump_getfreqi2c	Pump	To check the frequency of the pump (how fast it is pumping)	No. The frequency always comes out as zero, even when the pump is on.
pump_onoff	Pump	To turn the pump on	Yes.

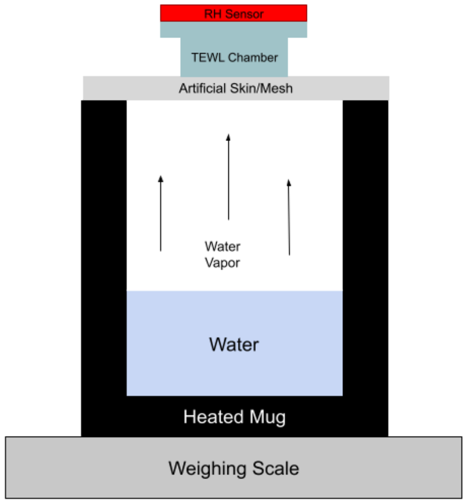
		and off using a digital write method – requires VCC pump pin to be connected to the VBUS seeeduino pin and the IOX pump pin to be connected to the D3 seeeduino pin.	
pump_testcode	Pump	To use I2C to connect to the pump and turn it on and off	No.
rh_sensor_bluetooth	RH sensor	Sending humidity readings from the RH_sensor to a phone via the LightBlue app.	Sort of. Humidity readings are taken and some value is sent via bluetooth, but it is sent via hex and is hard to tell if the number is really related to the RH reading at all.
temp_testcode	Thermocouple	Reading temp from the thermocouple	No. Code isn't fully developed yet
wetcup_testcode	RH sensor	Reads RH and temp from the Adafruit SHTC3 every 1.01-1.02 seconds and prints the time, temp, and RH to the serial monitor for easy data collection.	Yes

Table 1. TEWL device code files and descriptions

Calibration Methods

The team will calibrate the type-K thermocouple using water baths of different known temperatures. Any offset values will be taken into account for using the I2C code to allow for more accurate readings.

The team used a wet-cup method [2] to calibrate and validate readings of the RH sensor. This experiment is meant to generate a calibration curve that maps readings from the RH sensor to a corresponding TEWL value in g/(m^2*h). This calibration curve is also meant to account for the RH sensor's response time (τ=8s [3]) and any potential lag from the Bluetooth data transmission.



a.

b.

Figure 3. a. Diagram of the setup of the wet-cup method.
b. Pictures of the wet-cup method setup.

The setup consists of a heated mug with an adjustable temperature setting ranging from 105-150F, a weighing scale that measures up to 1/100 of a gram, a layer of artificial skin (stainless steel mesh), the TEWL chamber, and the RH sensor. The stainless steel mesh acts as a semi-permeable membrane to mimic the water transport rate through the outer layer of human skin, and then the TEWL chamber and RH sensor are placed on top to take RH measurements.

The mug is filled partially with water and then heated up to different temperatures between 110F and 150F to generate different rates of water vapor flux. Both the RH readings and total weight of water in the mug are recorded over time. Then, water vapor flux can be calculated by taking the total amount of water lost and dividing it by the area of the mug's opening and time. By repeating this for multiple temperatures of water, RH and water vapor flux can be plotted against each other to generate a calibration curve for all future RH readings.

Data Analysis

Data Analysis of the TEWL measuring device begins by taking the recorded sensor values and running it through the MATLAB files. Using the calibration curve from the wet-cup method, RH values obtained from the device are converted into TEWL values in MATLAB.

Results

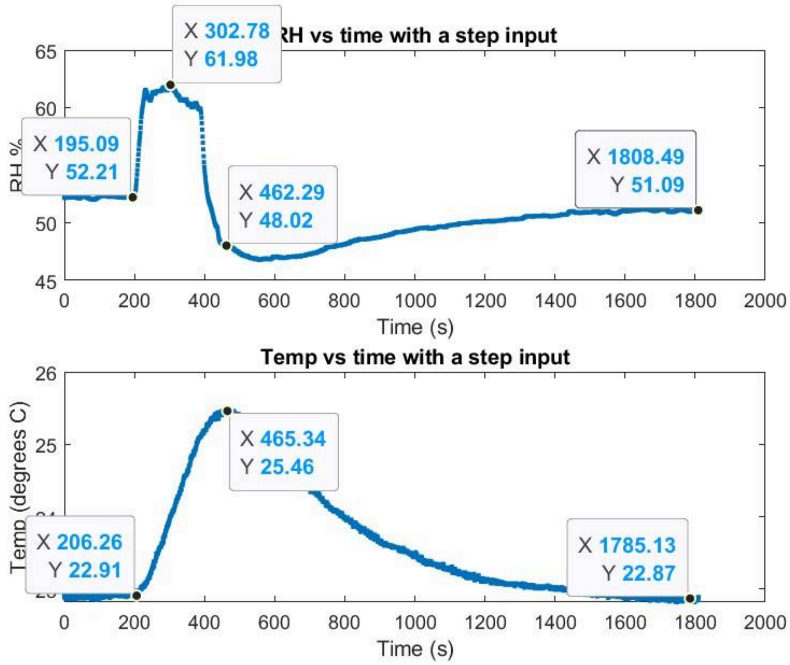


Figure 4. Plotted relative humidity and temperature from the RH sensor. The chamber and sensor were placed on skin at 195s and then taken off at 385s.

During preliminary testing of the RH sensor, it was found that the RH sensor's built-in temperature sensor has a response time of >80s, which is too long for the lab's skin stretch tests. This is because the time constant changes with temperature and τ=8s is only valid at 25°C [3]. Since RH measurements are calculated using the measurements from the built-in temperature sensor, and there is no way to separate moisture readings from temperature measurements, the team is currently searching for an alternative sensor to use for the device.

Future Work

The TEWL-measuring device will be used during *in vivo* experiments with George to characterize the subject's skin's barrier function throughout the duration of the experiment. The convenience and user-friendliness of the device should be evaluated during testing, and further adjustments should be made accordingly.

Alternative options should be considered for the pump. The team may try more temperature and strength settings for the hair dryer, or potentially 3D-print an attachment to funnel the air into the chamber. The team also needs a new sensor to calculate humidity, and is considering using an absolute humidity sensor since its measurements are not directly dependent on temperature.

The PLA 3D-printed casing for the device could be altered so that it is more aesthetically pleasing, less bulky, and allows for more user comfort. Additionally, double-sided tape or a rubber gasket could be added to the lip of the TEWL chamber opening so that it sits flush against the skin without any gaps.

References

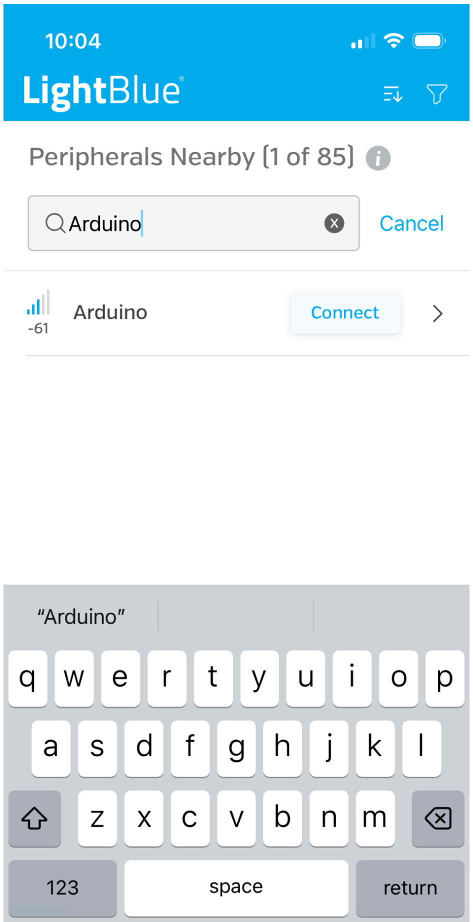
[1] A. D. Sivakumar, R. Sharma, C. Thota, D. Ding, and X. Fan, “WASP: Wearable Analytical Skin Probe for Dynamic Monitoring of Transepidermal Water Loss,” ACS Sensors, vol. 8, no. 11, pp. 4407–4416, Nov. 2023, doi: 10.1021/acssensors.3c01936.

[2] J. Nuutinen, E. Alanen, P. Autio, M. R. Lahtinen, I. Harvima, and T. Lahtinen, “A closed unventilated chamber for the measurement of transepidermal water loss,” Skin Research and Technology, vol. 9, no. 2, pp. 85–89, Apr. 2003, doi: 10.1034/j.1600-0846.2003.00025.x.

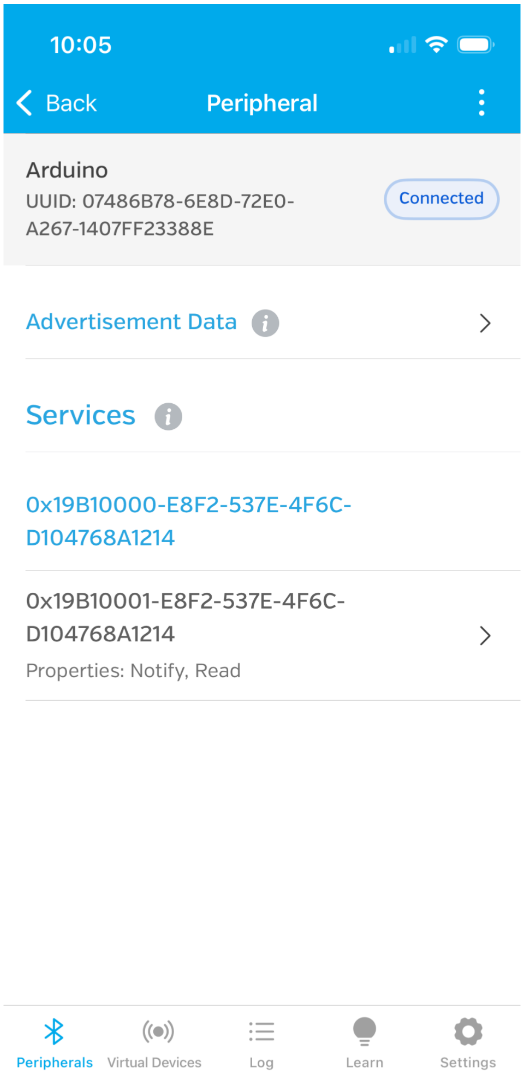
[3] “Datasheet SHTC3,” Accessed: July 27, 2026. [Online]. Available: https://cdn.sparkfun.com/assets/1/1/f/3/b/Sensirion_Humidity_Sensors_SHTC3_Datasheet.pdf

Appendix A: Connecting the Seeeduino to LightBlue

1. Download the LightBlue app
2. Upload code with Bluetooth connectivity instructions to the Seeeduino (e.g. rh_sensor_bluetooth)
3. Open the LightBlue app
4. In the search bar, look for “Arduino”

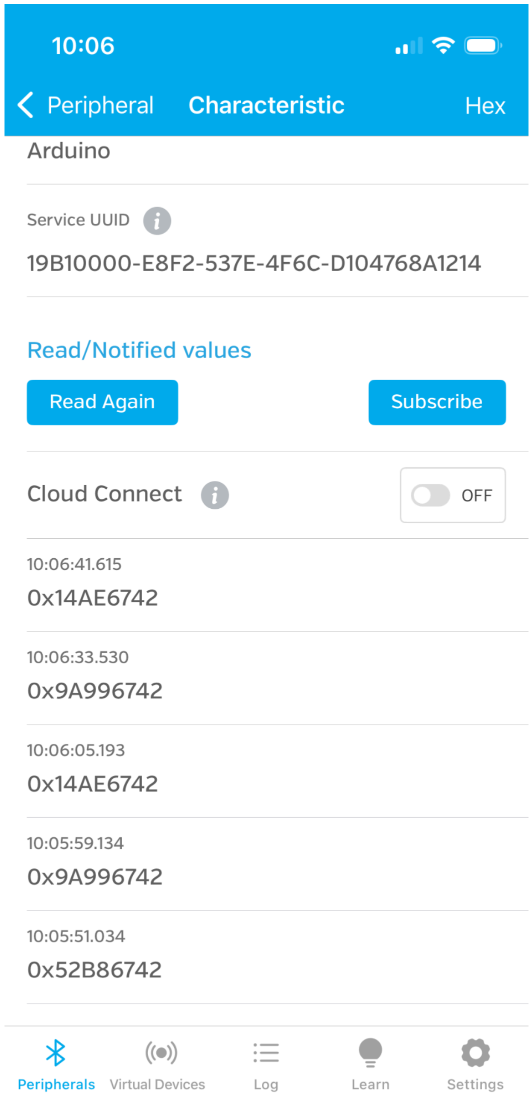
    a. 

5. Hit the connect button and wait for the Arduino to connect. It should look like the image below once connected.



a.

6. Click on the arrow next to “Properties: Notify, Read” (above)
7. Read values sent from the Seeeduino by clicking the “Read Again” button in the below image.



a.

8. For more information, go to https://wiki.seeedstudio.com/XIAO-BLE-Sense-Bluetooth-Usage/

Bill of Materials

Part Name	Quantity	Cost ($)	Manufacturer	Part Number	Supplier
Type-K Thermocouple	1	9.95	Adrafruit	270	DigiKey
Relative Humidity Sensor	1	13.11	SparkFun	SEN-16467	SparkFun
Qwicc Cable (Long)	1	2.75	SparkFun	PRT-17257	SparkFun
Qwicc Cable (Short)	1	1.95	SparkFun	PRT-17258	SparkFun
Qwicc Adapter	1	1.60	SparkFun	DEV-14495	SparkFun
MCP9600 I2C Thermocouple Amplifier	1	15.95	Adrafruit	4101	DigiKey
MCU (XIAO nRF52840a)	1	9.99	seeedstudio	102010448	seeedstudio
3.7V Li-Po Battery	1	7.99	Grepow Inc.	3254-GRP443535-1C-3.7V-450MAHWITHPCM-ND	DigiKey
4410 Battery Charging Board	1	5.95	Adrafruit	1528-4410-ND	DigiKey
One way valve	1	5.26	McMaster	6079T54	McMaster
Female Pigtail Cable	1	5.21	CompuCablePlusUSA	5412-DC-01501F-ND	DigiKey
Male Pigtail Cable	1	5.21	CompuCablePlusUSA	5412-DC-01501M-ND	DigiKey
Male Pins	4	3.25	3M	929400E-01-36-ND	DigiKey
Female Pins	4	4.17	3M	929974E-01-36-ND	DigiKey
Battery Cable	1	0.75	Adrafruit	1528-261-ND	DigiKey
Screw Terminal Adapter	1	27	XaviTech	4454-SCREWTERMINALADAPTER-ND	DigiKey
Micropump	1	213	XaviTech	4454-V100-5	DigiKey

				V-ND	
Rubber Sheet	1	6.68	McMaster		McMaster
Tube	1	0	Engineering Stock Room	N/A	Engineering Stock Room
Protoboard	1	0	Engineering Stock Room	N/A	Engineering Stock Room
Molecular Sieve 4A	1	0	Lape Lab	N/A	Lape Lab
PLA	1	0	MakerSpace	N/A	MakerSpace
Resin	1	0	MakerSpace	N/A	MakerSpace
Cotton ball	1	0	MakerSpace	N/A	MakerSpace
Woven Wire Mesh	1	9.89	uxcell	B08XZ2HSG8	Amazon
Smart Heated Mug	1	53.99	Armygo	B0H2VTY86X	Amazon